Course: World of Computing Gr 8 26

Lesson 2.4: Hardware

https://codehs.com/lms/assignment/93877828/lesson_plan

Description	What is hardware? How does hardware work? In this lesson, hardware is broken down into the different physical components of computers and how they contribute to the function of the computer as a whole.
Objective	Students will be able to: • Understand and identify the physical components of a computer & their roles in computer functionality
Activities	2.4.1 Video: Hardware 2.4.2 Check for Understanding: Hardware Quiz 2.4.3 Check for Understanding: Pick the Label 2.4.4 Free Response: Label Your Computer 2.4.5 Free Response: Computer Analogy 2.4.6 Free Response: Hardware vs. Software
Prior Knowledge	• basic components of a computer system and how they are organized • understand and identify different types of software and their functions
Planning Notes	• Review the slides and the exercises in the lesson. • Sticky notes will be needed for this activity (or paper and tape). • The *Simulate a Computer* handout required significant setup. Refer to this handout for more information. • There is a handout that accompanies this lesson. It can be used as an in-class activity or a homework assignment. Determine how and if this handout will be used and make the appropriate number of printouts prior to the class period.
Standards Addressed	**CSTA 3A Standards** **Name** **Description**

3A-CS-02	Compare levels of abstraction and interactions between application software, system software, and hardware layers.

England 3 Standards

Name	Description
ENG.3.5	understand the hardware and software components that make up computer systems, and how they communicate with one another and with other systems

PRAXIS 5652 Standards

Name	Description
IV.A.1.e	explain why binary numbers are fundamental to the operation of computer systems
IV.B.2	Be familiar with methods to store, manage, and manipulate data
V.A.1	Know that operating systems are programs that control and coordinate interactions between hardware and software components
V.A.1.a	identify hardware components and their functions
V.A.4	Be familiar with computers as layers of abstraction from hardware (e.g., logic gates, chips) to software (e.g., system software, applications)
V.A.4.a	identify appropriate abstraction layers for hardware and software components

Teaching and Learning Strategies

Lesson Opener:

- Have students brainstorm and write down answers to the discussion questions listed below. Students can work individually or in groups/pairs. Have them share their responses. [5 mins]

Activities:

- Watch the lesson video and take the corresponding quiz. This quiz is a quick check for understanding. [10 mins]
- Complete the *Pick the Label* quiz. [5 mins]
  - Encourage students to search and find another image of a hardware device. Share with the class and have the class label the found images.
- Complete the *Label Your Computer* free response activity. [10 mins]
  - Students may realize that hardware devices such as routers may be missing. Where do they think they are in the school? Do you know?
- Complete the *Computer Analogy* free response activity. [10 mins]
  - Encourage creativity. Students can even draw a diagram of the analogy if needed.
- Complete the *Hardware vs Software* free responses activity. [10 mins]
  - Play a "scattergories" game where a student shares one comparison and if other students have the same one, it doesn't count as a point. Points are only scored for comparisons that no one else mentioned!
- Complete the *Simulate a Computer* handout. [20 mins]
- Complete the *What Do Your Suggest?* handout. [15 mins]
  - This handout can also be used as a homework assignment.

Lesson Closer:

- Have students reflect and discuss their responses to the end of class discussion questions. [5 mins]

Discussion Questions

Beginning of Class:

- What are some of the major hardware devices that are used in a computing system?
  - *CPUs, monitors, keyboards, etc.*
- What is the job of these hardware devices?
  - *Answers will vary based on the last question.*
- How do the hardware devices communicate with the computer?
  - *The devices could be hardwired or use BlueTooth technology. Note that this is really just how the devices are connected - they communicate in binary/logic gates etc.*

End of Class:

- Once connected, how does the hardware device communicate with the computer?
  - *The device uses logic gates and binary code to communicate with the computer.*
- What are the two types of memory? Explain the difference between the two.
  - *Main Memory stores currently executing program instructions and external Memory is used to store data long-term.*

	• What is the difference between Random Access Memory and Read Only Memory? ◦ *Random Access Memory (RAM) will only store information if the power is on and stores currently running applications and corresponding data. Read Only Memory (ROM) will keep information stored regardless of if the power is on and is where BIOS is stored.*
Resources/Handouts	Simulate a Computer (student version) Simulate a Computer (teacher version) What Do You Suggest? (teacher) What Do You Suggest? (student)

Vocabulary

Term	Definition
Computer	A person or device that makes calculations, stores data, and executes instructions according to a program.
Hardware	The physical components of a computer

Modification: Advanced	Modification: Students Needing Additional Support	Modification: English Language Learners
	• Print out video slides for students to reference	• Print out video slides for students to reference